

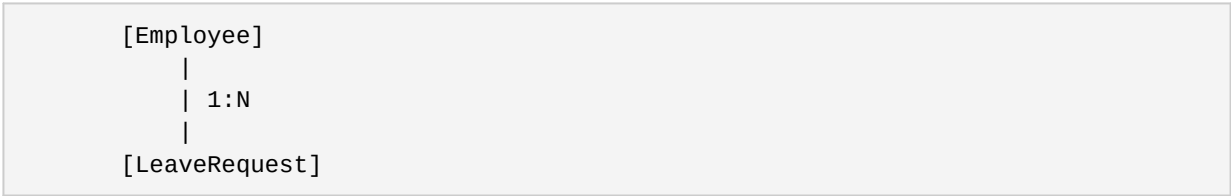
Data Model Documentation

Overview

The application manages the following core domains to facilitate the submission, approval, and tracking of employee leave requests within our organization.

- **Employee:** Responsible for managing employee data, including leave balances and requests. Cardinality: 1:N with LeaveRequest.
- **LeaveRequest:** Responsible for managing leave requests, including their status and associated dates. Cardinality: N:1 with Employee.

Entity Relationship Diagram



Entities

Employee

Purpose: Represents an employee within the organization, including their leave balance and requests.

Table/Collection Name: Employee

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier for the employee
name	VARCHAR	Yes	Max 255 chars	Employee's full name
email	VARCHAR	Yes	Max 255 chars, unique	Employee's email address
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time

Field Name	Type	Required	Constraints	Description
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Has many LeaveRequest (1:N): Inverse via LeaveRequest.employee_id

Validation Rules:

- email must be non-empty and unique across all records
- name must be non-empty

LeaveRequest

Purpose: Represents a leave request submitted by an employee, including its status and associated dates.

Table/Collection Name: LeaveRequest

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier for the leave request
employee_id	UUID	Yes	Foreign key to Employee.id	Employee who requested the leave
start_date	DATE	Yes	Must not be in the past	Start date of the leave
end_date	DATE	Yes	Must be after start_date	End date of the leave
status	VARCHAR	Yes	Must be one of 'Pending', 'Approved', 'Rejected'	Status of the leave request
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time

Field Name	Type	Required	Constraints	Description
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Employee (N:1): Foreign key employee_id references Employee . id

Validation Rules:

- start_date must be a valid date and not in the past
- end_date must be a valid date and after start_date
- status must be one of 'Pending', 'Approved', 'Rejected'

Relationships and Constraints

Foreign Keys

From Entity	Field	To Entity	Field	Cardinality	Delete Behavior
LeaveRequest	employee_id	Employee	id	N:1	CASCADE

Unique Constraints

- Employee(email): Ensures no duplicate email addresses

Data Validation Rules

Business Rules:

- LeaveRequest . start_date must be a valid date and not in the past: If violated, return error "Start date must not be in the past."
- LeaveRequest . end_date must be a valid date and after start_date: If violated, return error "End date must be after start date."
- LeaveRequest . status must be one of 'Pending', 'Approved', 'Rejected': If violated, return error "Invalid leave status."

Format Rules:

- `Employee.email`: Must match regex
`^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`

Data Lifecycle

Creation

- A new `LeaveRequest` is created when an employee submits a leave request.
- Default status is `Pending`
- `created_at` is automatically set to the current timestamp

Updates

- `LeaveRequest.status` can only be updated by an authorized user (e.g., manager)
- `updated_at` is automatically updated on any modification

Archival / Soft Delete

- Records are not archived but can be marked as deleted by setting the status to `'Rejected'`

Indexes

- Primary key: `id`
- Foreign keys: `employee_id`
- Search fields: `name`, `email` (for common queries)

Example Records

Employee

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "John Doe",
  "email": "john.doe@example.com",
  "created_at": "2026-01-15T10:30:00Z",
  "updated_at": "2026-01-15T10:30:00Z"
}
```

LeaveRequest

```
{
  "id": "123e4567-e89b-12d3-a456-426614174000",
  "employee_id": "550e8400-e29b-41d4-a716-446655440000",
  "start_date": "2026-01-15",
  "end_date": "2026-01-20",
  "status": "Pending",
  "created_at": "2026-01-15T10:30:00Z",
  "updated_at": "2026-01-15T10:30:00Z"
}
```